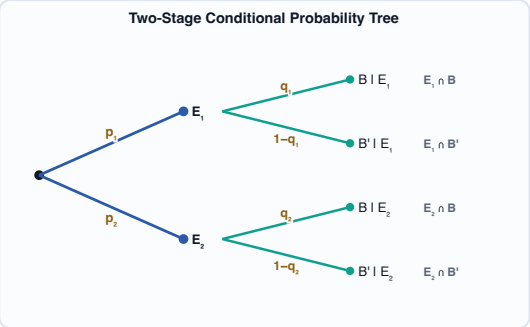
25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Conditional Probability & Bayes' Rule Practice Exercises

1 MCQ

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.10$. What is the probability of the complement class $P(\neg C_1)$?



A 0.90
B 0.95
C 0.05
D 1.10

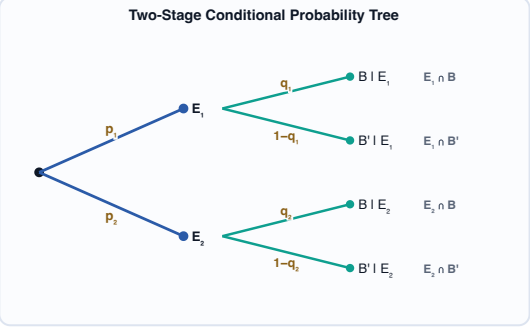
2 MCQ

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.20$. What is the probability of the complement class $P(\neg C_1)$?

A 0.80
B 0.90
C 0.10
D 1.20

3 MCQ

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.60$. What is the probability of the complement class $P(\neg C_1)$?



A 0.40
B 0.70
C 0.30
D 1.60

4 MCQ

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.05$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

A 0.0450

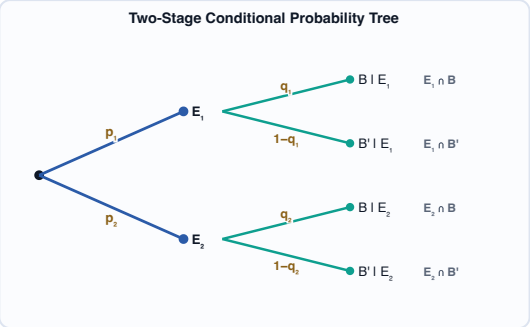
B 0.9500

C 0.8500

D 0.0050

5 MCQ

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.10$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.



A 0.0900

B 1.0000

C 0.8000

D 0.0100

6 MCQ

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.15$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

A 0.1350

B 1.0500

C 0.7500

D 0.0150

7 MCQ

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.20$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

A 0.1800

B 1.1000

C 0.7000

D 0.0200

8 MCQ

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.25$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

A 0.2250

B 1.1500

C 0.6500

D 0.0250

9 MCQ

An Ordinary Least Squares linear model $\hat{p}(t) = 2t + 11$ is evaluated at $t = 3$. Determine the predicted response $\hat{p}(3)$.

A 19
B 21
C 6
D 17

10 MCQ

Considering a dataset, the prior probability of class C_1 is $P(C_1) = 0.20$. What is the probability of the complement class $P(\neg C_1)$?

A 0.90
B 0.10
C 1.20
D 0.80

11 MCQ

In gradient descent on loss $h(u)$, parameter $u_0 = 44$ is updated with learning rate $\eta = 0.1$ and gradient $\nabla h(u_0) = 70$. Determine the updated parameter $u_1 = u_0 - \eta \nabla h(u_0)$.

A 61
B 37
C 30
D 41

12 MCQ

Examining a dataset, the prior probability of class C_1 is $P(C_1) = 0.60$. What is the probability of the complement class $P(\neg C_1)$?

A 0.70
B 0.40
C 0.30
D 1.60

13 MCQ

A medical test for a rare disease has a 99\% accuracy rate in detecting the disease when it is present (True Positive Rate) and a 95\% accuracy rate in identifying individuals without the disease (True Negative Rate). If the prevalence of the disease in the population is 0.5\%, what is the probability that a person who tests positive actually has the disease?

A Approximately 8.7\%
B Approximately 1.0\%
C Approximately 9.1\%
D Approximately 0.5\%

14 MCQ

A medical diagnostic test for a rare disease is designed to be highly sensitive and specific. The disease affects 1 in every 1000 individuals in the population. The test has a sensitivity of 99% (meaning it correctly identifies 99% of individuals who have the disease) and a specificity of 95% (meaning it correctly identifies 95% of individuals who do not have the disease). If a randomly selected individual from the population tests positive, what is the probability that they actually have the disease?

A. Approximately 0.019

B. Approximately 0.99

C. Approximately 0.05

D. Approximately 0.95

15 MCQ

In an applied problem, a machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.15$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

A. 1.0500

B. 0.1350

C. 0.7500

D. 0.0150

16 MCQ

Suppose a machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.20$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

A. 1.1000

B. 0.7000

C. 0.0200

D. 0.1800

17 MCQ

An observed machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.25$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

A. 0.0250

B. 1.1500

C. 0.6500

D. 0.2250

18 MCQ

Consider a machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.30$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

A. 0.2700

B. 1.2000

C. 0.6000

D. 0.0300

19 MCQ

In anomaly detection, anomaly prior $P(A) = 0.02$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|\neg A) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly?

A 0.279

B 0.140

C 0.950

D 0.391

20 MCQ

Considering anomaly detection, anomaly prior $P(A) = 0.04$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|\neg A) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly?

A 0.221

B 0.442

C 0.950

D 0.619

21 MCQ

Examining anomaly detection, anomaly prior $P(A) = 0.06$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|\neg A) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly?

A 0.950

B 0.767

C 0.274

D 0.548

22 MCQ

For the configuration in anomaly detection, anomaly prior $P(A) = 0.08$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|\neg A) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly?

A 0.872

B 0.311

C 0.950

D 0.623

23 MCQ

Analyzing anomaly detection, anomaly prior $P(A) = 0.10$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|\neg A) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly?

A 0.340

B 0.950

C 0.679

D 0.951

24 MCQ

Within anomaly detection, anomaly prior $P(A) = 0.12$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|\neg A) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly?

A 0.361

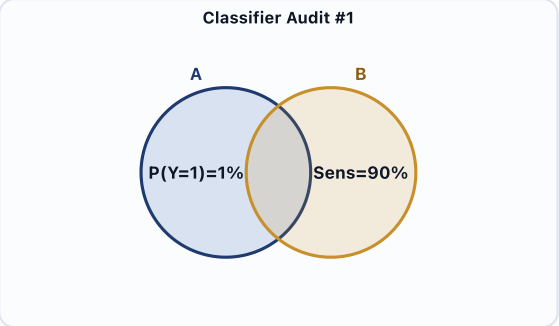
B 0.950

C 1.011

D 0.722

25 MCQ

In classifier audit #1, positive-class prevalence is $P(Y = 1) = 1\%$, sensitivity is $P(\hat{Y} = 1 \mid Y = 1) = 90\%$, and false-positive rate is $P(\hat{Y} = 1 \mid Y = 0) = 5\%$. By Bayes' Rule, find the precision $P(Y = 1 \mid \hat{Y} = 1)$.



A $\frac{9}{10}$

B $\frac{9}{1000}$

C $\frac{11}{13}$

D $\frac{2}{13}$